

EDA REPORT

RECRUITMENT EFFICIENCY IMPROVED

GROUP 3

TRI MEDIAN

Struktur DataFrame

Coloumn	Data Type
reqruitment_id	int64
departement	object
job_title	object
num_applicants	int64
time_to_hire_days	int64
cost_per_hire	float64
source	object
offer_acceptance_rate	float64

Describe Data

	recruitment_id	num_applicants	time_to_hire_days	cost_per_hire	offer_acceptance_rate
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000
mean	2500.500000	155.621000	47.191400	5214.826236	0.650832
std	1443.520003	84.164264	23.864934	2730.999185	0.202052
min	1.000000	10.000000	7.000000	507.160000	0.300000
25%	1250.750000	83.000000	26.000000	2820.597500	0.480000
50%	2500.500000	157.000000	47.000000	5218.290000	0.650000
75%	3750.250000	229.000000	67.000000	7611.412500	0.830000
max	5000.000000	299.000000	89.000000	9998.910000	1.000000

The dataset exhibits high variability in applicant volume, hiring duration, and recruitment costs, indicating differences in process efficiency and potential relationships among variables that influence recruitment success.

Check Missing Value & Duplicate

1

The dataset contains no missing values in any of the variables.

2

The dataset does not contain any duplicated data.

```
df.isnull().sum()
```

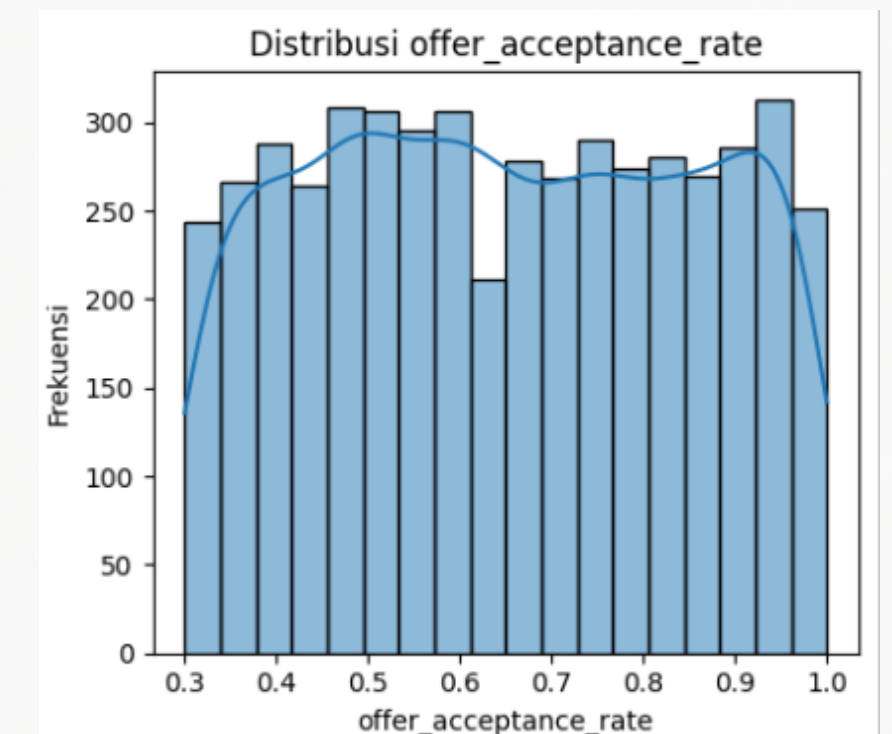
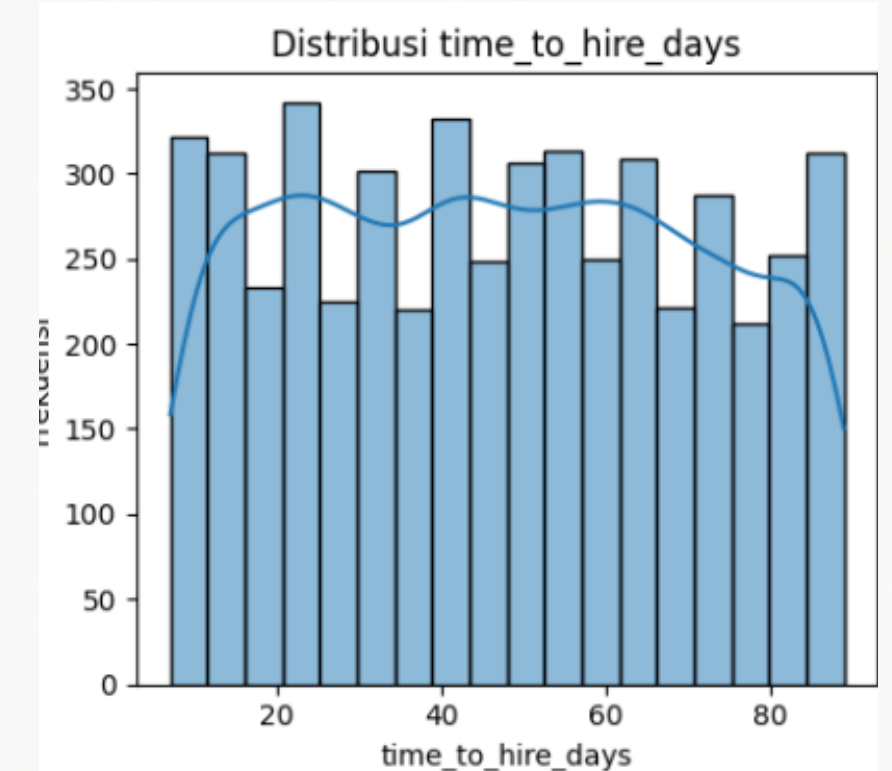
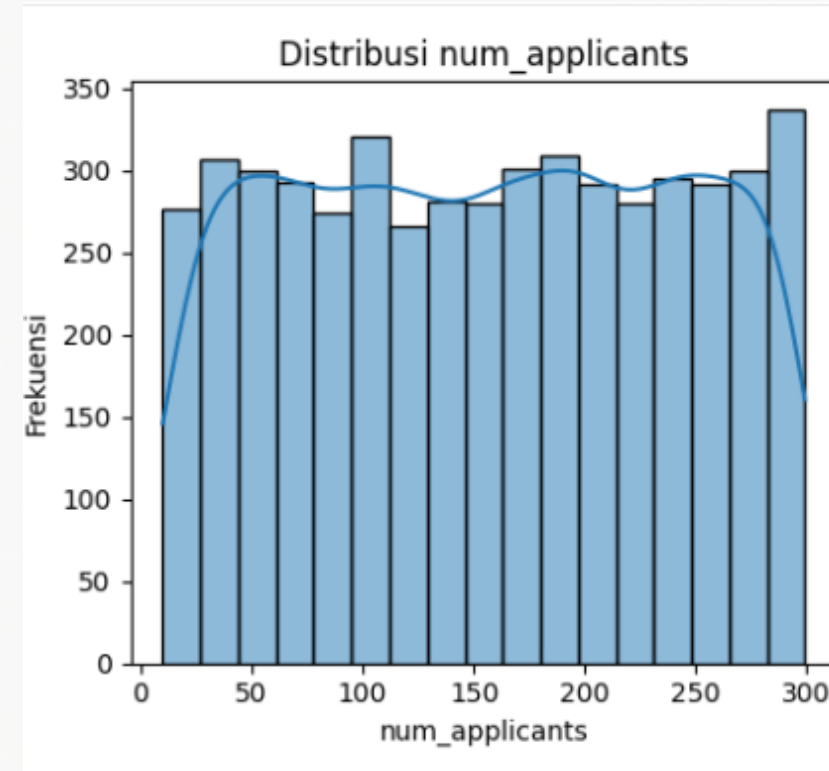
...	0
recruitment_id	0
department	0
job_title	0
num_applicants	0
time_to_hire_days	0
cost_per_hire	0
source	0
offer_acceptance_rate	0

```
df.duplicated().sum()  
... np.int64(0)
```

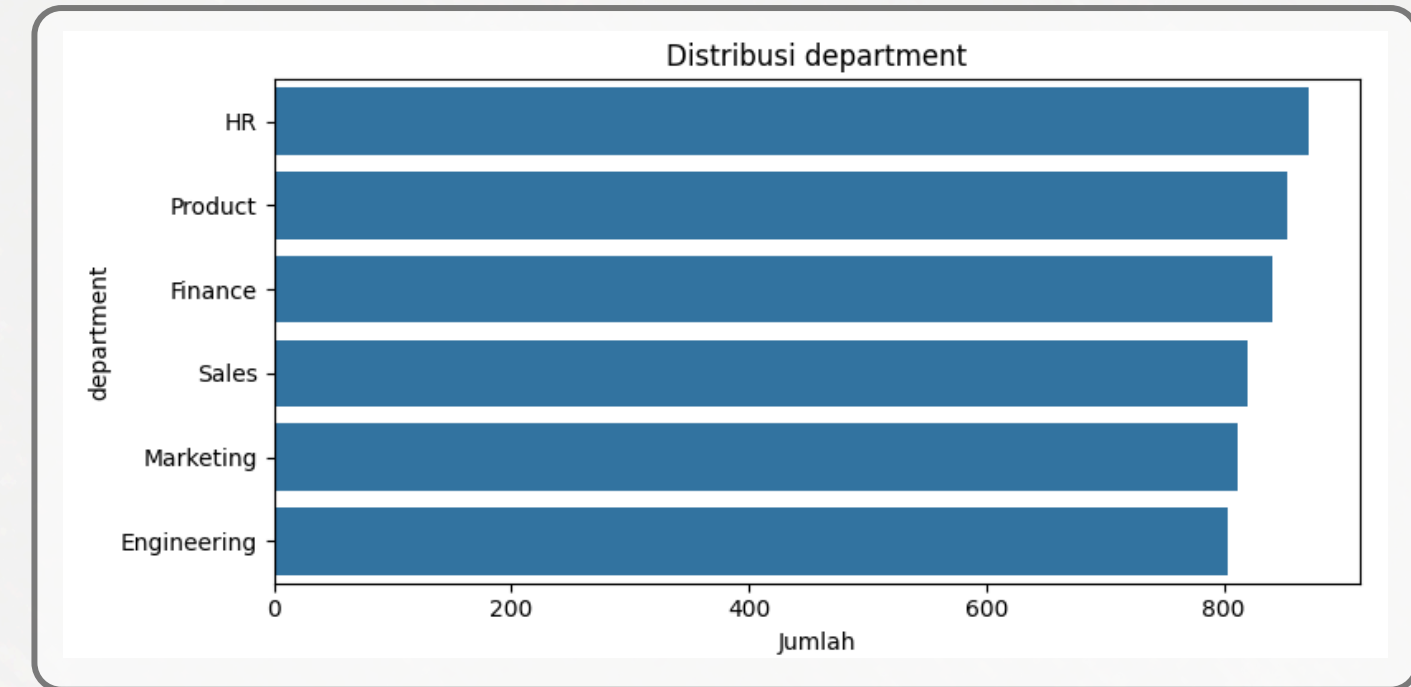

Univariate Numeric

- 1 The distribution of num_applicants is approximately normal, with no significant outliers detected
- 2 The time_to_hire_days variable shows an approximately normal distribution and no significant outliers.
- 3 The distribution of cost_per_hire is close to normal, and no significant outliers are observed
- 4 The offer_acceptance_rate variable follows an approximately normal distribution with no significant outliers detected

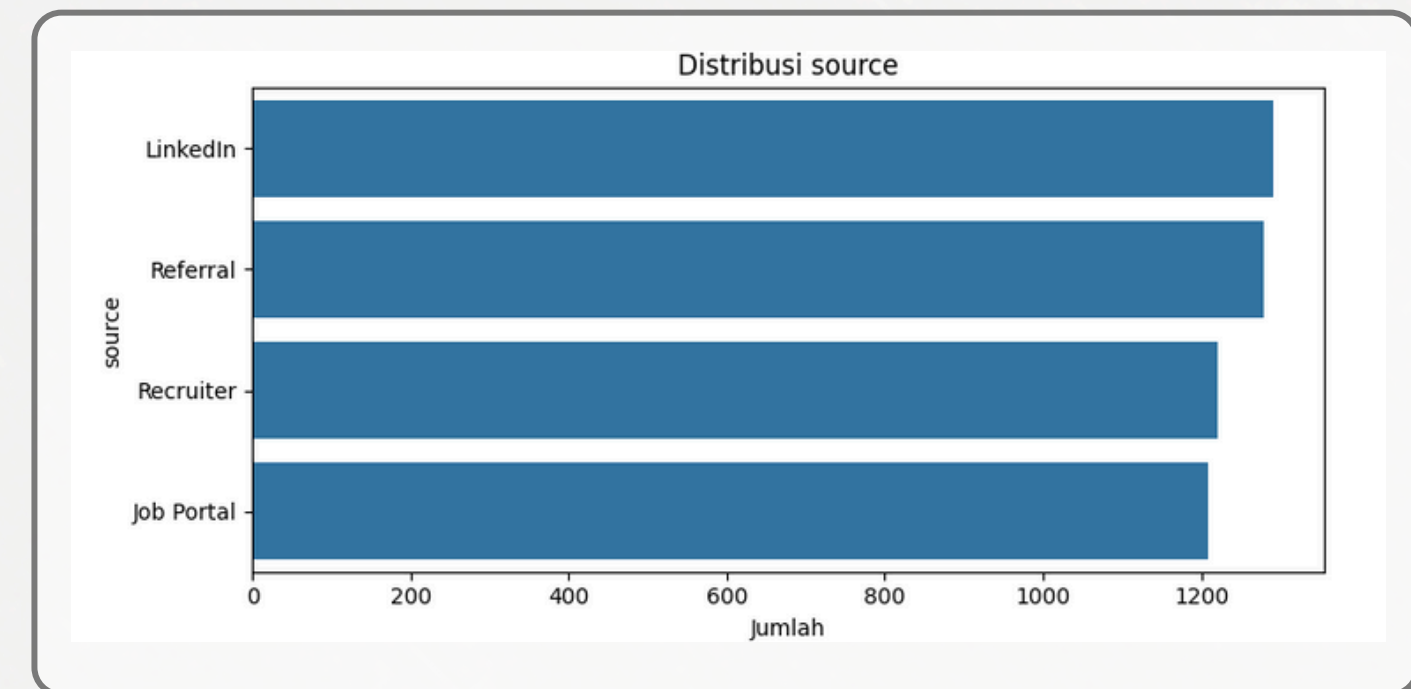
EDA REPORT



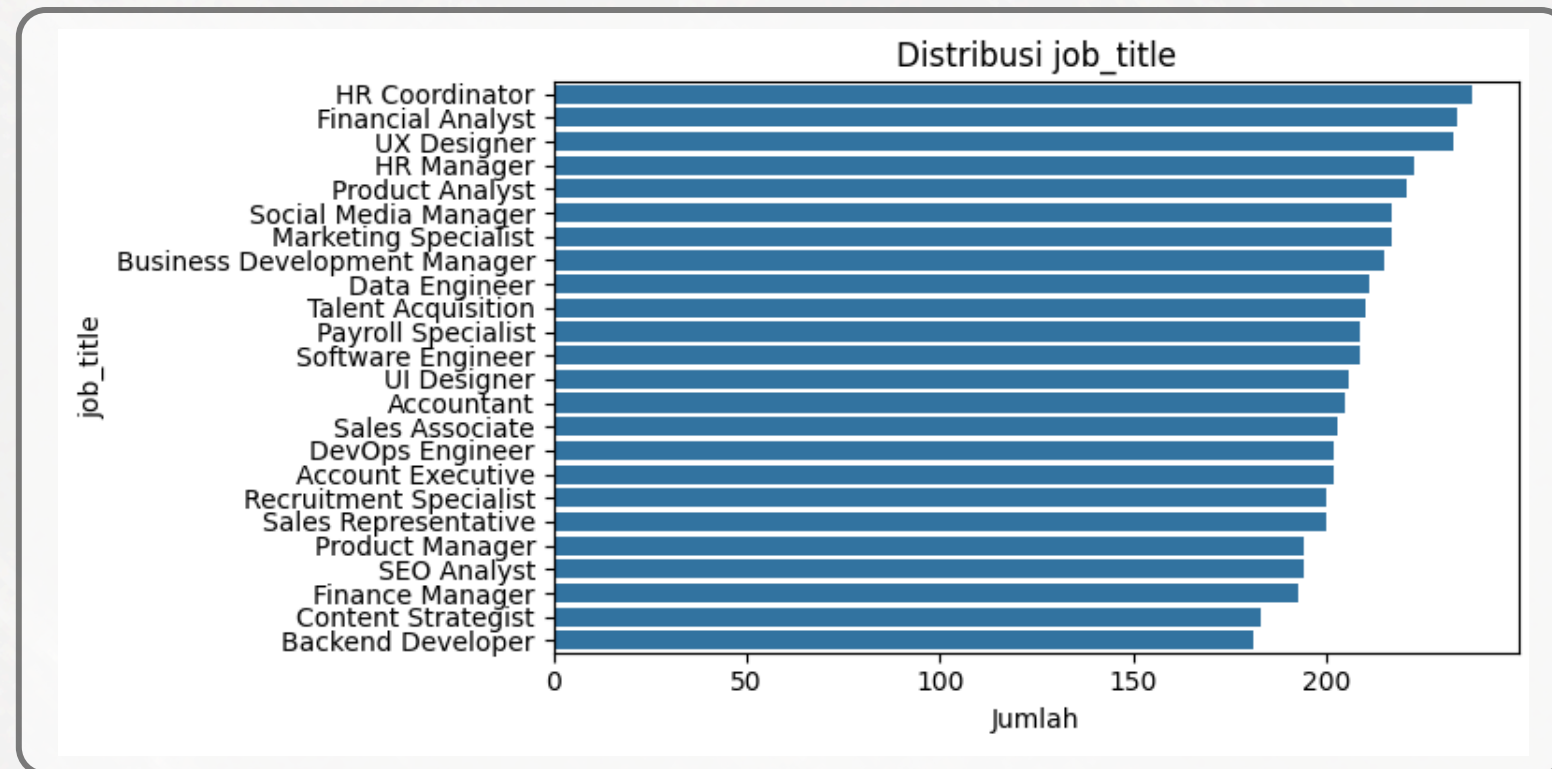
Univariate Categorical



- 1 The department variable has six unique categories with relatively even frequencies and no rare categories detected.

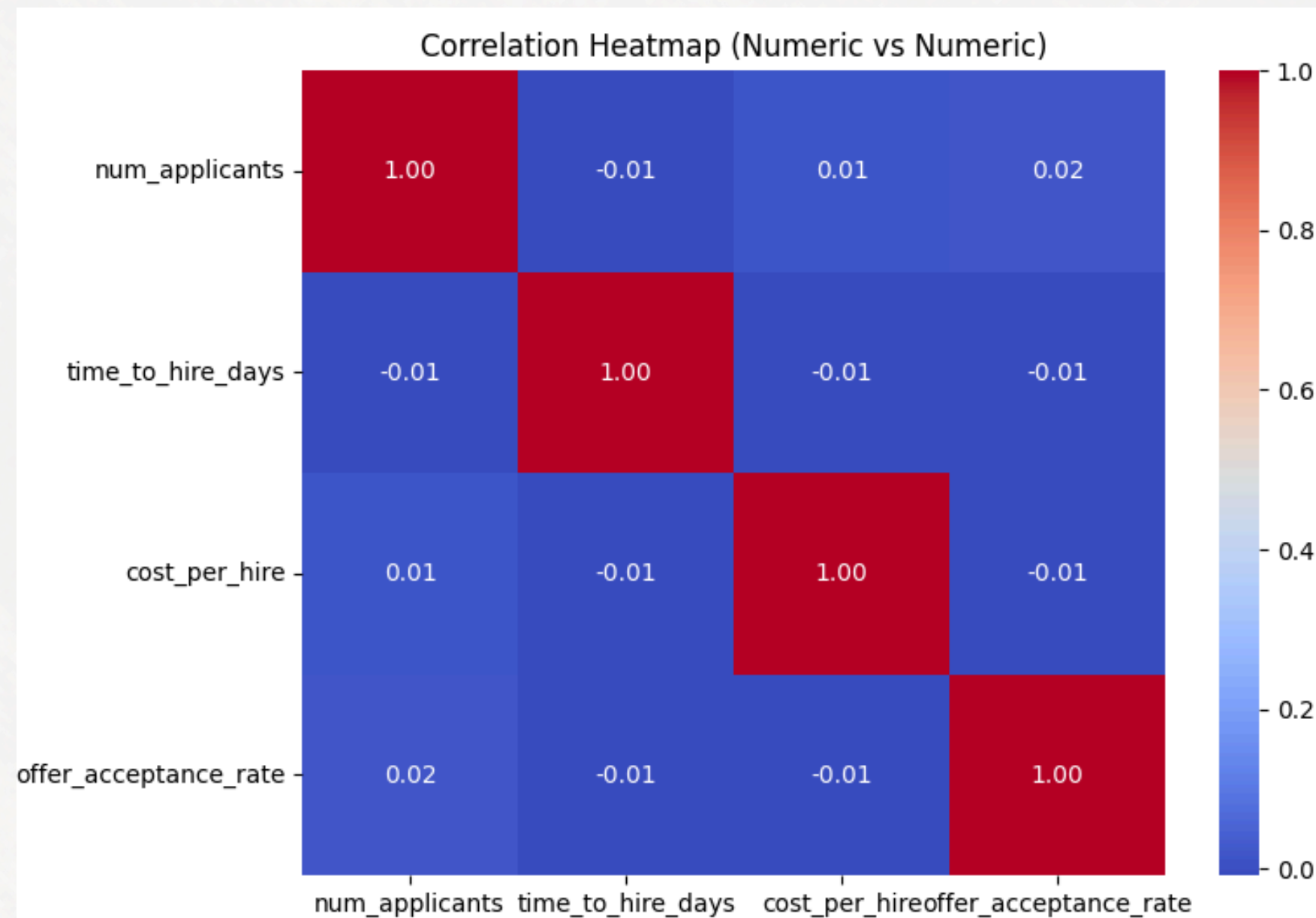


- 2 The source variable has four unique categories with relatively even frequencies and no rare categories detected.



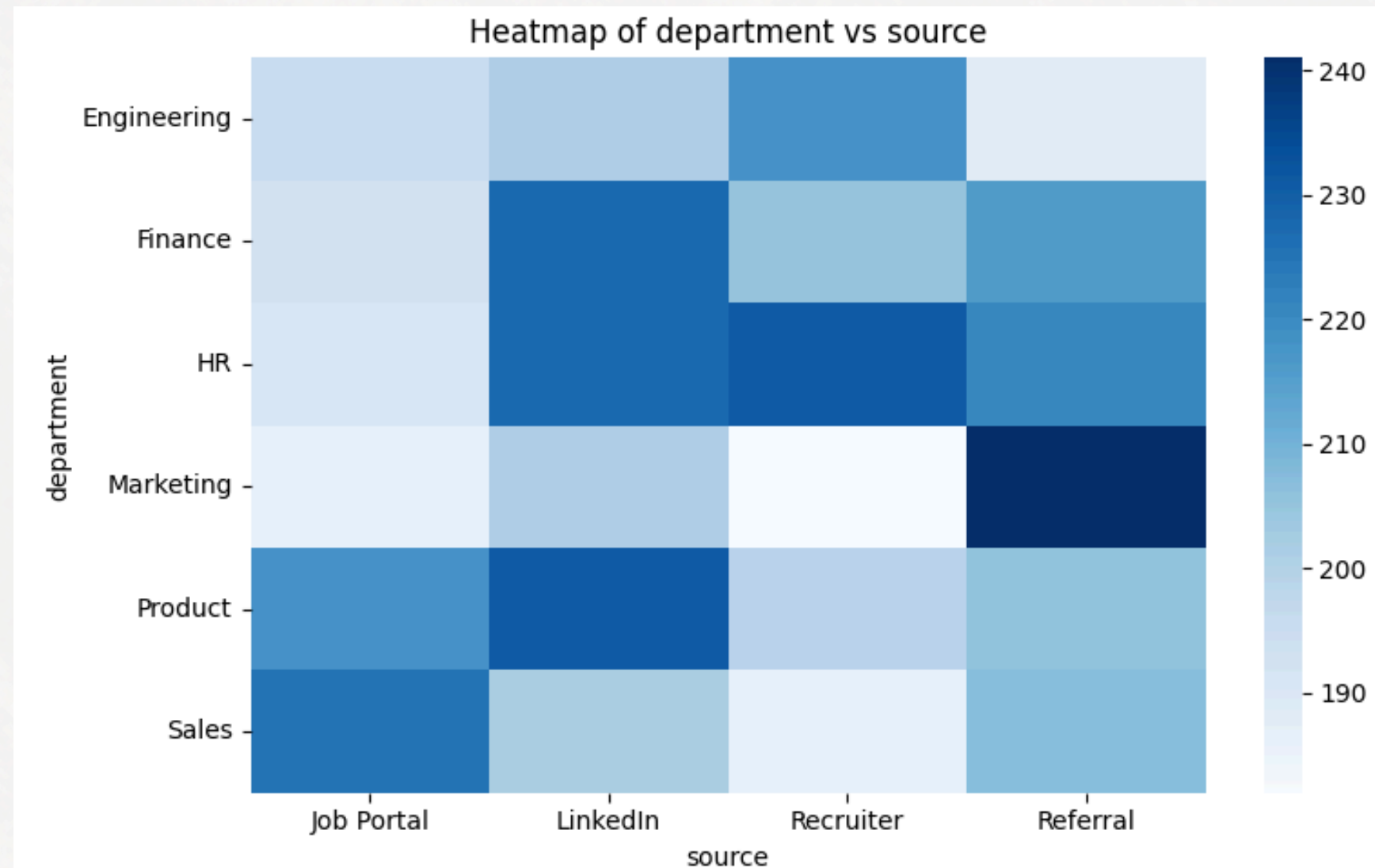
- 3 The job_title variable has 24 unique categories with relatively even frequencies and no rare categories detected.

Correlation Heatmap



Overall, the data indicates that there is no significant linear relationship (correlation) among the numerical variables in this recruitment process. Correlation values approaching zero (-0.01 to 0.02) suggest that each metric functions as an independent variable.

Corelation Heatmap



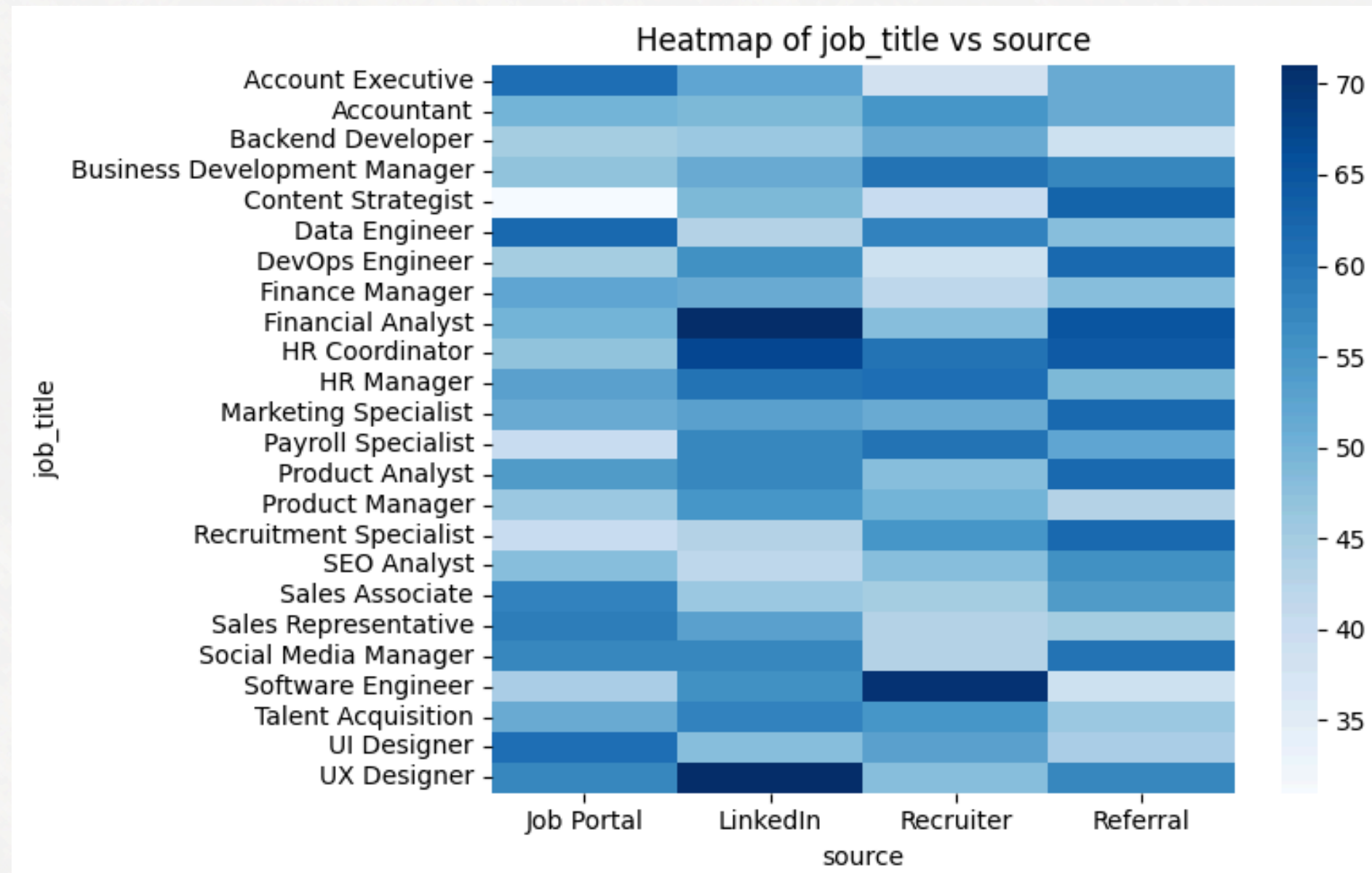
INSIGHT :

- LinkedIn and Job Portal are the primary recruitment sources across most departments.
- HR and Product show a relatively high reliance on Recruiters.
- Marketing has the highest contribution from Referrals compared to other departments.
- Sales relies more on Job Portals and LinkedIn than on Recruiters.

CONCLUSION :

Sourcing strategies vary across departments, indicating that recruitment channels are tailored to the specific workforce needs of each department.

Corelation Heatmap



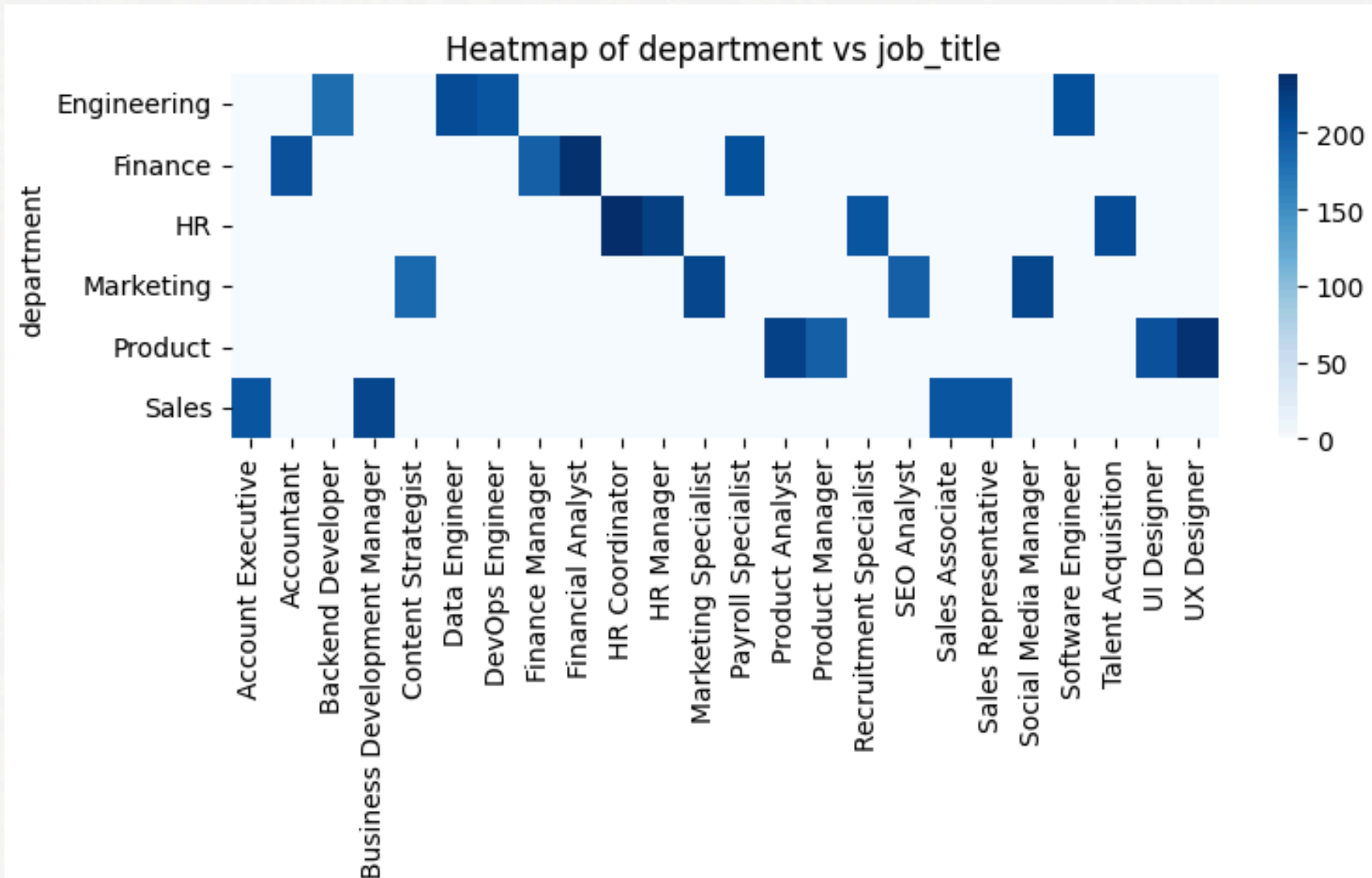
INSIGHT :

- Job Portals are dominant for operational and entry-mid level positions such as Sales Associate, Account Executive, and Content Strategist.
- LinkedIn is highly effective for professional and specialized roles, including Software Engineer, Data Engineer, HR Manager, and UX Designer.
- Recruiters are widely used for strategic or highly specialized positions, such as Finance Manager, Recruitment Specialist, and Product Manager.
- Referrals contribute significantly to certain roles, particularly in Marketing, Product, and selected technical positions.

CONCLUSION :

Each job title exhibits a distinct recruitment source pattern, reflecting the effectiveness of specific channels for different types of roles.

Corelation Heatmap



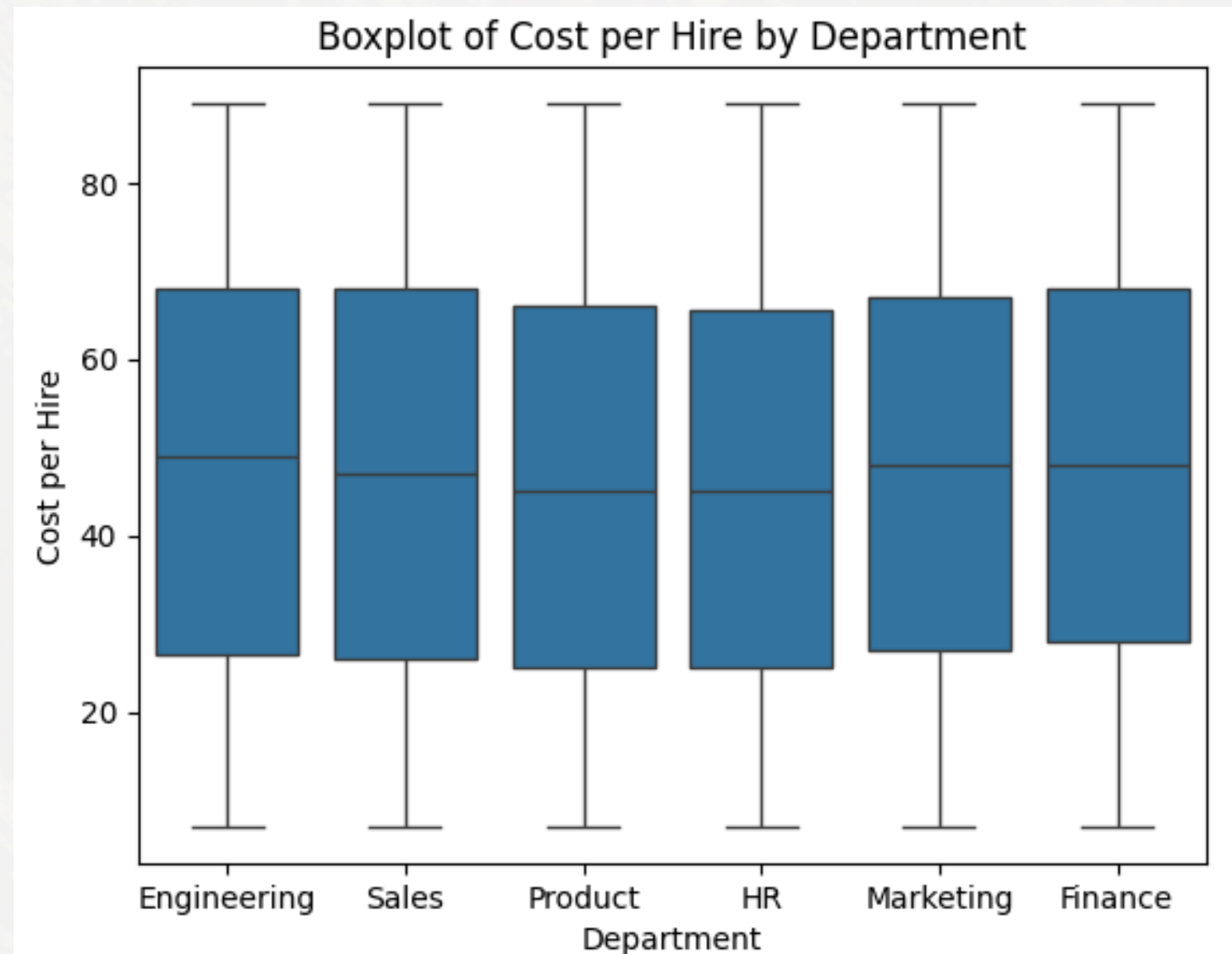
INSIGHT :

- Engineering is dominated by technical roles such as Backend Developer, Data Engineer, DevOps Engineer, Software Engineer, and UI/UX Designer.
- Finance is primarily associated with roles such as Accountant, Financial Analyst, and Finance Manager.
- HR is concentrated on roles including HR Coordinator, HR Manager, and Talent Acquisition.
- Marketing is dominated by roles such as Marketing Specialist, Social Media Manager, Content Strategist, and SEO Analyst.

CONCLUSION :

Job titles are logically distributed and consistently aligned with the functions of each department, with no indications of improper placement.

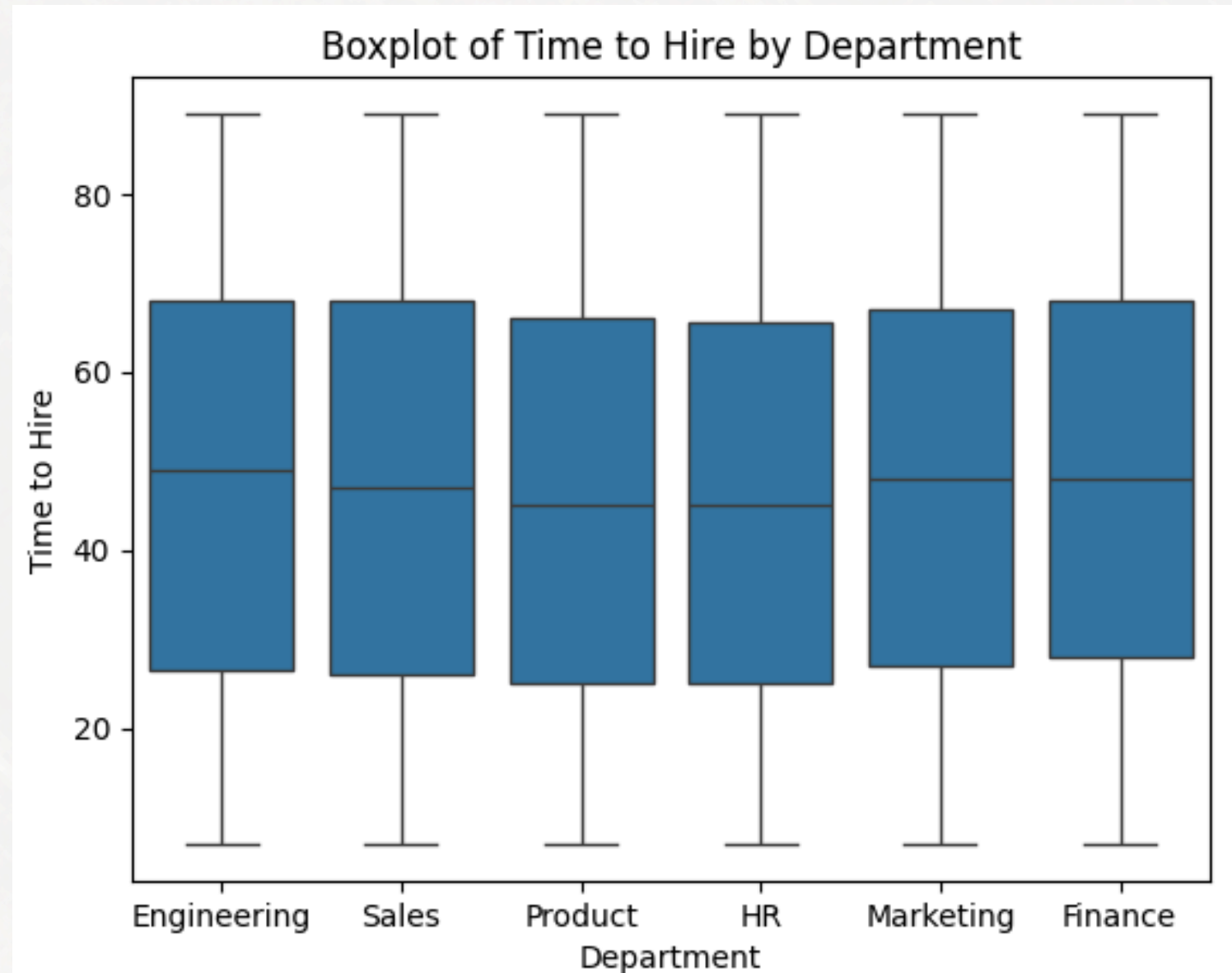
Boxplot Numeric and Categorical



INSIGHT:

- The cost per hire is relatively consistent across all departments, with similar median values.
- Product and Marketing show slightly higher median cost per hire compared to other departments.
- No department exhibits extreme values or unusually high variability in recruitment costs.

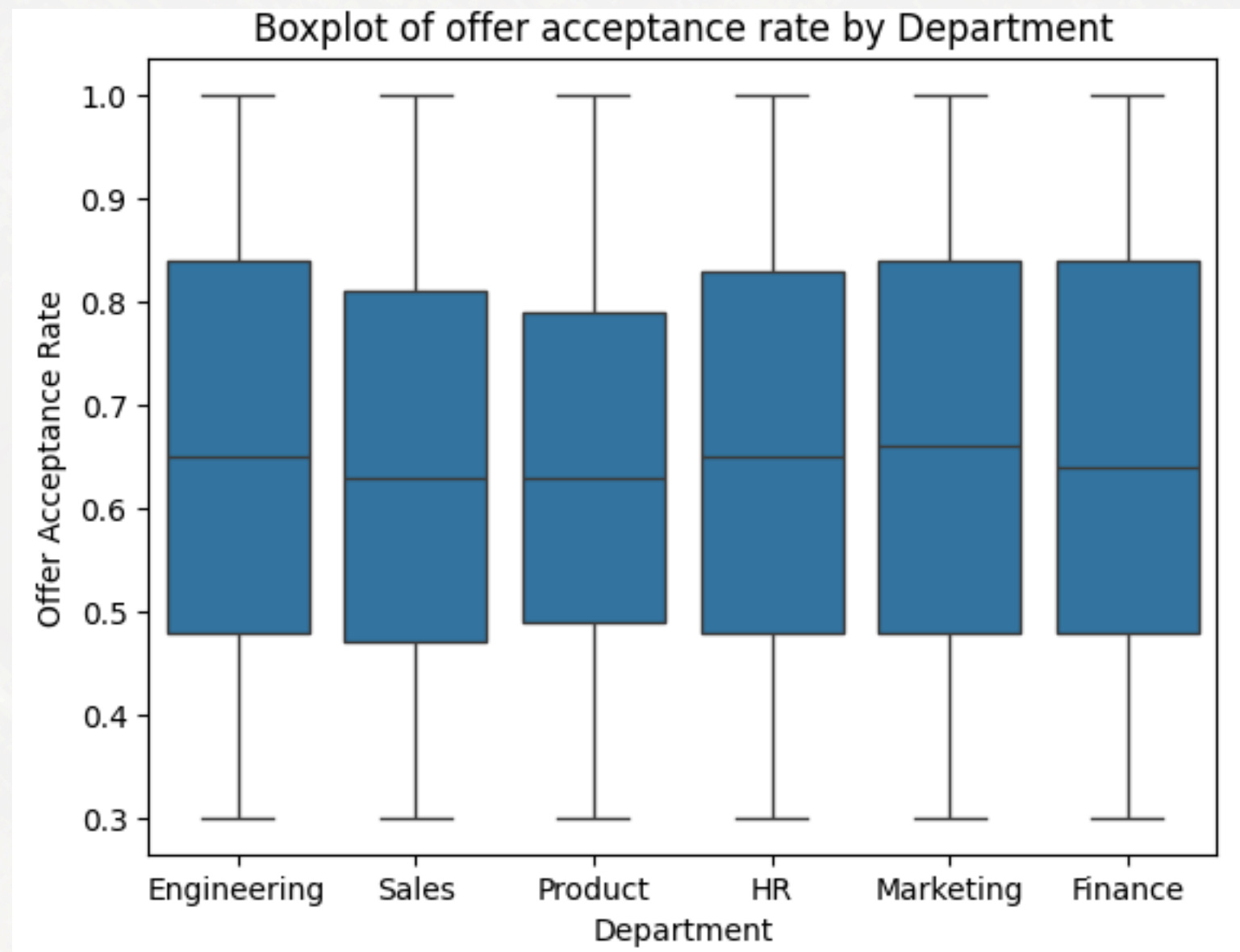
Boxplot Numeric and Categorical



INSIGHT :

- The Time to Hire distribution is highly consistent across all departments, indicating no significant differences in hiring duration between Engineering, Sales, Product, HR, Marketing, and Finance
- All departments exhibit similar median Time to Hire values (around 45–50 days) with comparable interquartile ranges (IQR), suggesting uniform hiring speed and variability.
- The similar overall range of Time to Hire across departments implies a well-standardized recruitment process, with no department-specific time bottlenecks observed.

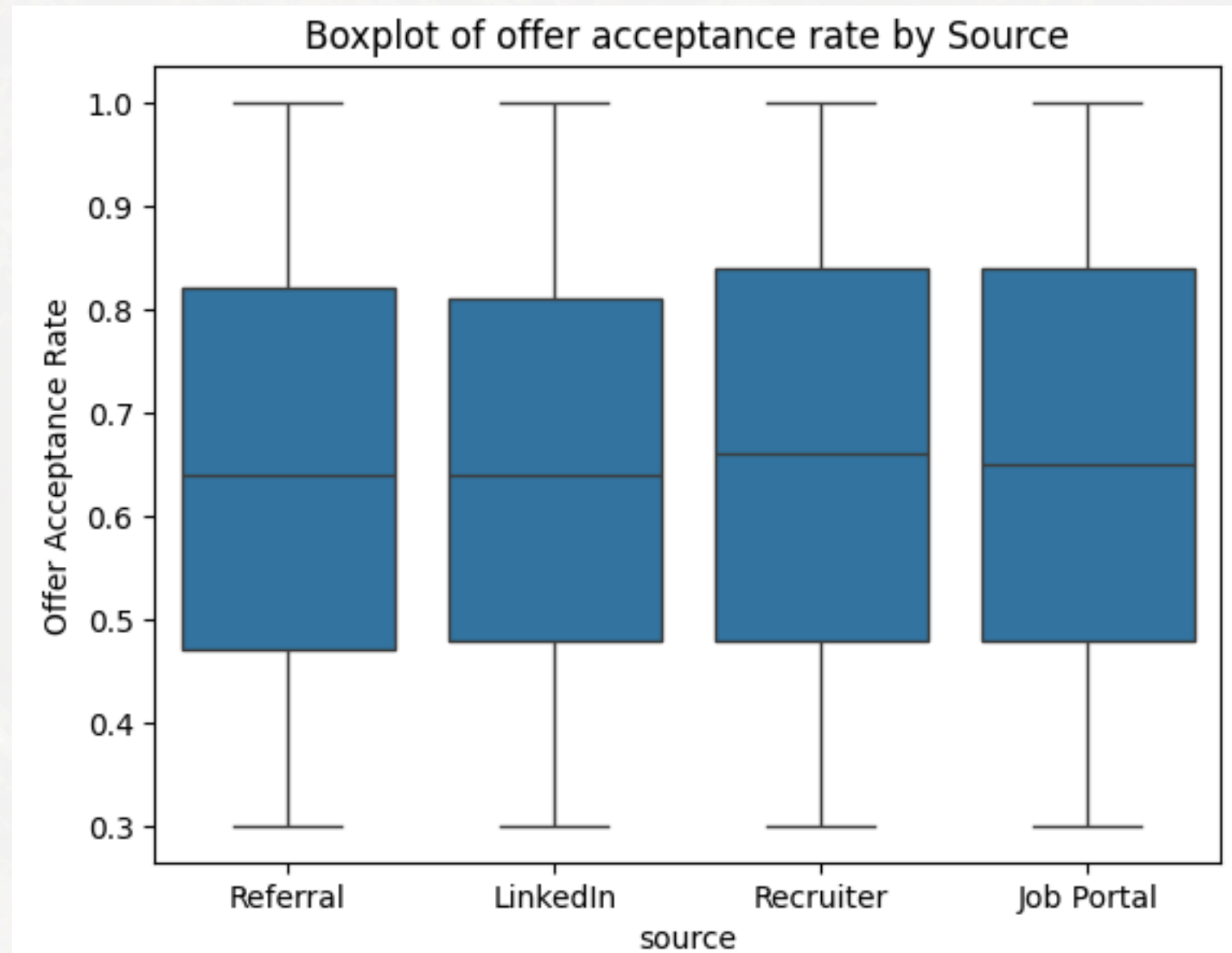
Boxplot Numeric and Categorical



INSIGHT :

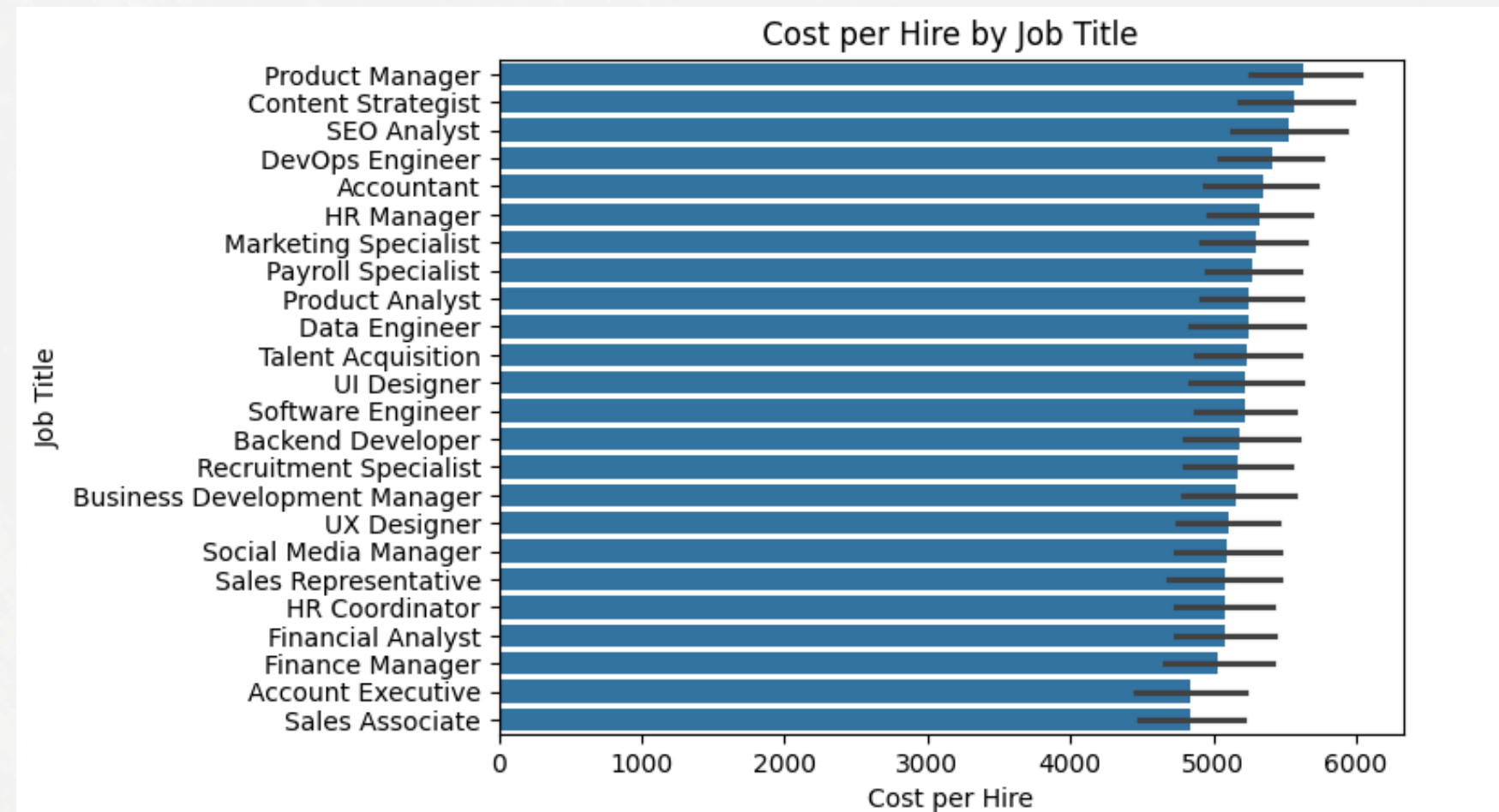
- The median Offer Acceptance Rate is consistent across departments, ranging from approximately 63% to 66, indicating similar effectiveness of offer strategies and employer attractiveness.
- Despite similar medians, all departments exhibit a wide acceptance rate range (around 30% to 100%), suggesting significant fluctuations across hiring periods or role types.
- No department stands out as a clear positive outlier, implying that improving offer acceptance is a company-wide challenge rather than a department-specific issue.

Boxplot Numeric and Categorical



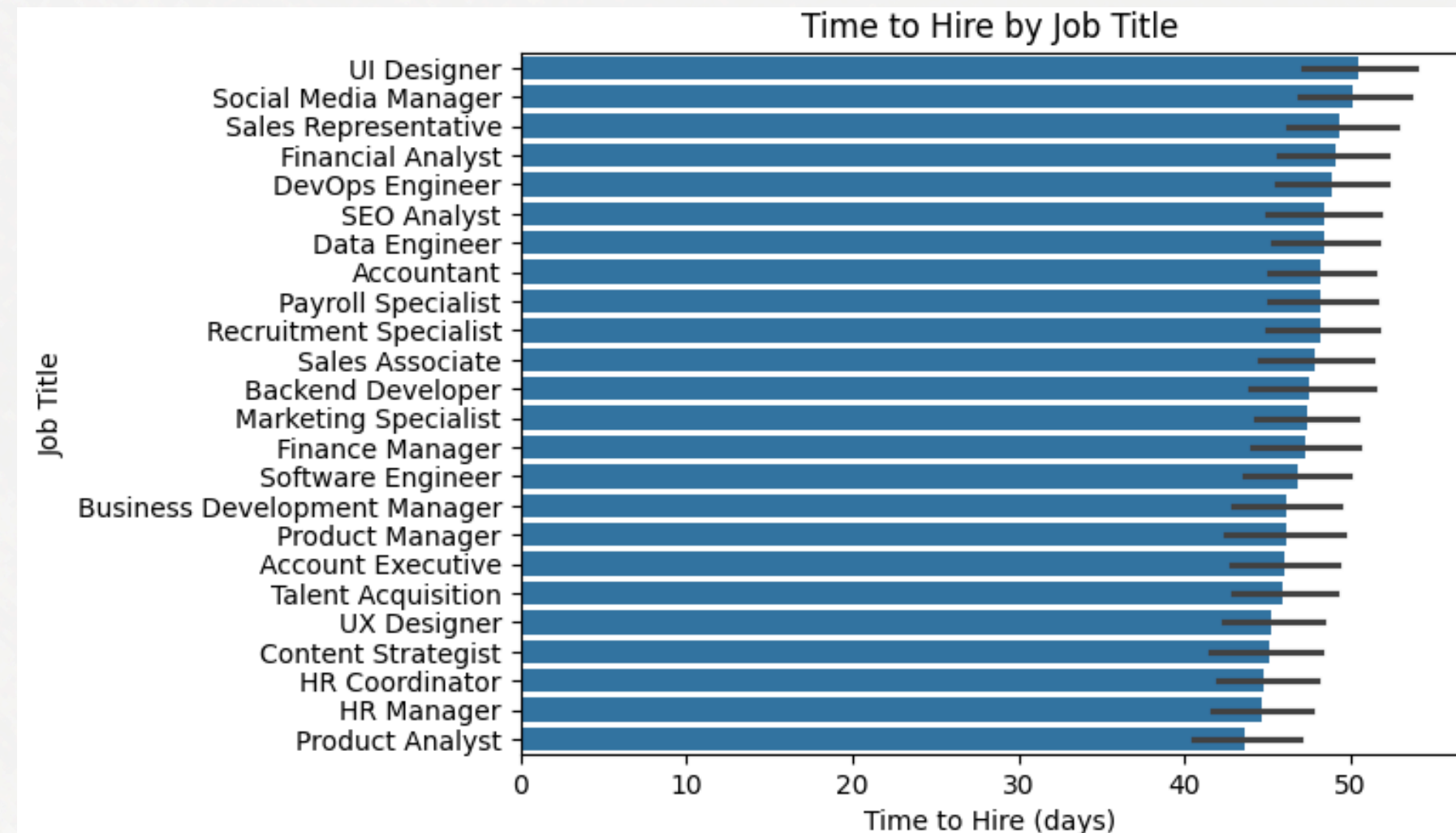
The box plot shows that offer acceptance rates are relatively similar across all recruitment sources (Referral, LinkedIn, Recruiter, and Job Portal). The similar median values indicate that no recruitment source significantly outperforms others in terms of offer acceptance. However, the wide range of values suggests that candidate responses vary, meaning that some candidates are more likely to accept offers while others are more likely to decline, even within the same source.

Barplot Numeric and Categorical



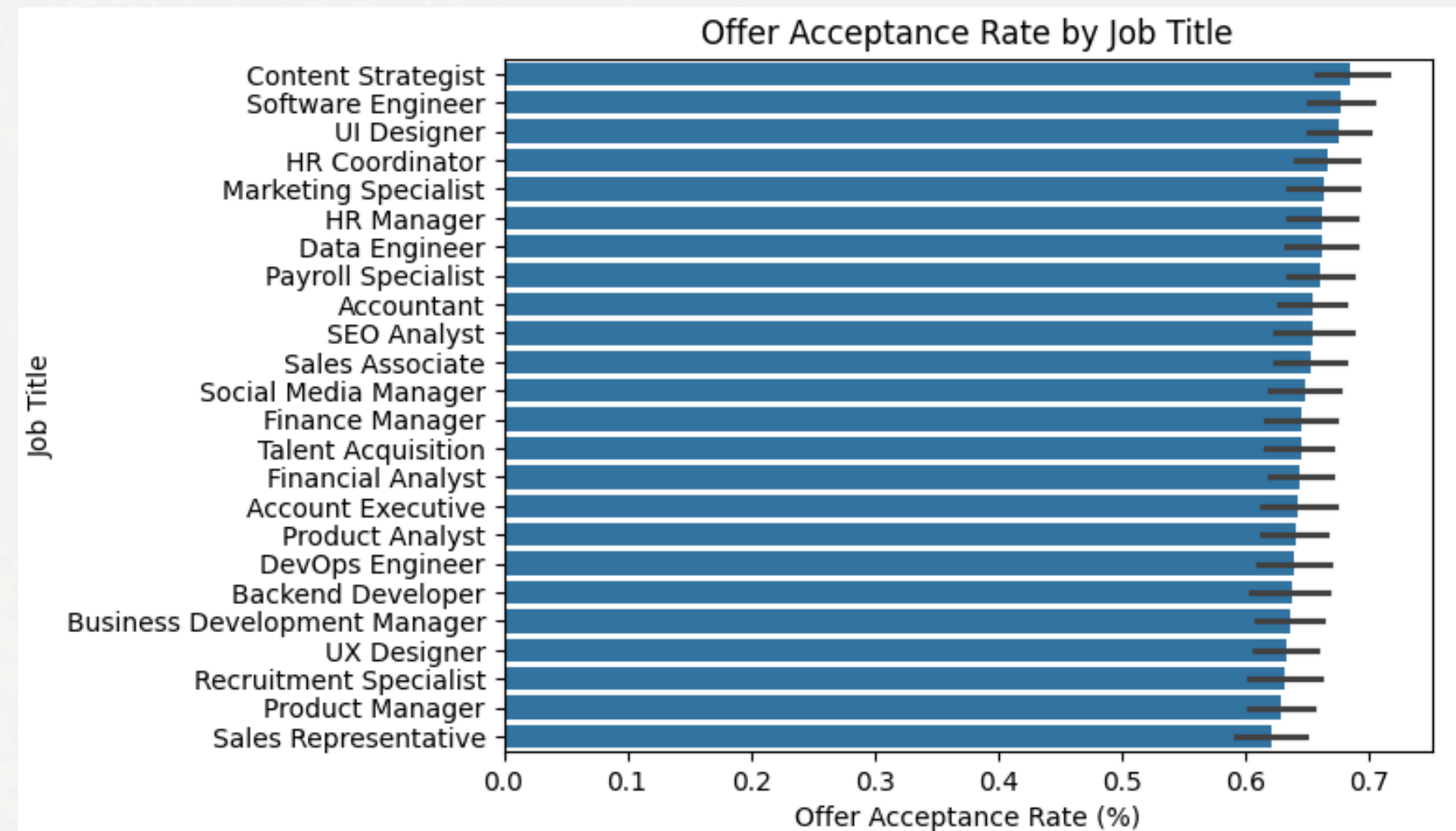
The analysis indicates that strategic and technical roles tend to have a higher cost per hire compared to operational roles. This suggests higher skill requirements and stronger competition for talent in these positions.

Barplot Numeric and Categorical



Certain roles such as UI Designer and Social Media Manager exhibit longer time to hire, while HR and analyst positions are filled more quickly. This reflects differences in selection complexity and candidate availability across roles.

Barplot Numeric and Categorical



Offer acceptance rates are relatively consistent across job titles, with certain roles showing higher acceptance than others. This indicates variation in offer attractiveness and labor market conditions by role.

Feature Engineering

acceptance_category	time_to_hire_category	applicant_pressure_index	cost_bucket
Very Likely Accept	Fast	11.666667	Low Cost
Likely Accept	Fast	37.428571	Low Cost
Likely Accept	Slow	0.144737	Medium Cost
Likely Reject	Fast	9.153846	Medium Cost
Likely Accept	Medium	7.432432	Medium Cost

- *Acceptance Category groups offer acceptance rates into discrete categories to simplify the interpretation of recruitment success.*
- *Time-to-Hire Category classifies hiring duration into fast, medium, and slow to assess recruitment process efficiency..*
- *Cost-per-Hire Bucket segments recruitment costs into low, medium, and high groups based on data distribution to identify cost efficiency differences.*
- *Applicant Pressure Index measures recruitment intensity by combining applicant volume and hiring duration to represent selection pressure.*

Data Standarization

```
# Copy Data
df_base = df2.copy()

# Drop Recruitment ID
df_base = df_base.drop(columns=["recruitment_id"])

# Standarisasi Coloumn Categorical
cat_cols = ["job_title", "department", "source"]

for col in cat_cols:
    df_base[col] = df_base[col].str.strip().str.lower()
```

Drop Recruitment ID and Standarization Coloumn Categorical

```
# Convert Category Menjadi Angka agar mudah dibaca model
target_map = {
    "likely reject": 0,
    "uncertain": 1,
    "likely accept": 2,
    "very likely accept": 3
}

df_base["target"] = (
    df_base["acceptance_category"]
    .str.lower()
    .map(target_map)
)
```

Convert offer category to be numeric

```
# Drop Coloumn Leakage/redundant

df_base = df_base.drop(columns=[
    "acceptance_category",      # turunan target
    "offer_acceptance_rate"     # sumber target
])
```

Drop Coloumn Leakage/redundant

Data Cleaning Model 1

Scaling Feature Numeric

```
df_3 = df_base.copy()

# Scaling fitur numeric
from sklearn.preprocessing import StandardScaler

num_cols = [
    "num_applicants",
    "time_to_hire_days",
    "cost_per_hire",
    "applicant_pressure_index"
]

scaler = StandardScaler()
df_3[num_cols] = scaler.fit_transform(df_3[num_cols])
```

One hot Encoding

```
# One Hot Encoding
cat_cols_lr = [
    "department",
    "source",
    "job_title",
    "time_to_hire_category",
    "cost_bucket"
]

df_3 = pd.get_dummies(
    df_3,
    columns=cat_cols_lr,
    drop_first=True
)
```


Data Cleaning Model 1

```
▶ # output result DF3 For Logistic Regression  
  
X_lr = df_3.drop(columns=["target"])  
y_lr = df_3["target"]
```

Preprocessing for Logistic Regression is more rigorous because the model is sensitive to feature scaling and requires standardized numerical data and reduced multicollinearity for stable and interpretable coefficient estimation.

Data Cleaning Model 2

One Hot Encoding

```
df_4 = df_base.copy()

# Label Encoding
from sklearn.preprocessing import LabelEncoder

cat_cols_rf = [
    "department",
    "source",
    "job_title",
    "time_to_hire_category",
    "cost_bucket"
]

label_encoders = {}

for col in cat_cols_rf:
    le = LabelEncoder()
    df_4[col] = le.fit_transform(df_4[col])
    label_encoders[col] = le
```

```
# output Result DF4 For Random Forest
X_rf = df_4.drop(columns=["target"])
y_rf = df_4["target"]
```

Preprocessing for Random Forest is kept minimal because the model is scale-invariant and can handle numerically encoded categorical features without assuming linear relationships.